

CASE STUDY

Garage Management System

Rebuilding workshop operations around the job card

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Case study prepared for portfolio purposes. Delivered under NDA with Petromin Express. Outcome figures are projections modelled from observed task-completion times, not measured production results.

THE PROBLEM

Disconnected systems slowed every part of workshop operations

Legacy tools forced teams to work across multiple systems, creating delays, duplicate work, and limited operational visibility.

DISCOVERY

Understanding the business

To understand the daily challenges, we observed workshop operations, interviewed teams across departments, and mapped how work moved from vehicle check-in to billing.

Interviewed the people running it

Station, finance and area managers, across roles and across sites.

Reviewed existing workflows

How a vehicle actually moved through a workshop, step by step.

Mapped the dependencies

System integrations and approval chains, and where each one stalled.

FINDINGS

Where the legacy system failed

Six issues recurred at every station we visited.

Teams worked across disconnected systems

Service, inventory and finance operated independently, requiring manual coordination.

Approvals slowed daily operations

Routine approvals delayed work, especially during busy hours.

Managers lacked real-time visibility

Operational updates were only available after work had already progressed.

Technicians switched between tools

Essential information was spread across multiple systems, interrupting servicing.

Inventory records were out of sync

Parts usage and job cards required manual reconciliation.

Peak hours bottlenecked

Approval queues and manual processes slowed workshop throughput.

WHAT TEAMS TOLD US

Five people, one problem, five descriptions of it

"I manually compare records across different systems every day. When something doesn't match, I have no way to know which one is correct."

"Every vehicle requires checking another system. Switching back and forth slows me down and increases mistakes."

"I oversee 12 stations, but I only learn about problems after they've already happened."

"Reconciling invoices with ZATCA is manual, and mismatches are often discovered days later."

"Employee information exists in multiple systems, so we maintain the same data twice."

Every role described the same fracture from a different side of it. Nobody was missing information - it existed. It just never arrived in one place, in time to act on.

PRODUCT ECOSYSTEM

A multi-role operational hierarchy

Designed as a connected hierarchy supporting multiple teams, workflows and responsibilities across workshop operations.

Area managers & auditors

Operational oversight, reporting, compliance monitoring.

Station managers

Operational monitoring, bay management, workflow visibility.

Service advisors

Vehicle check-in, customer coordination, approvals.

Technicians

Service execution, inspections, diagnostics, workflow updates.

Inventory teams

Stock tracking, parts requests, inventory transfers.

Finance teams

Invoices, payments, reconciliation, refunds.

DESIGN PRINCIPLES

Designing for operational speed and clarity

Five foundational decisions that guided every design and interaction across the platform.

Workflow-first UX

Interfaces were designed around operational states and guided workflows instead of traditional page-based navigation.

Low cognitive load

The experience minimised unnecessary interactions to support fast-paced workshop environments.

Scan-first interactions

VIN or plate scanning, quick actions and contextual inputs reduced manual entry effort and errors.

Real-time visibility

Operational states and workflow progress remained visible throughout the service lifecycle.

Role-based complexity

Each user experienced only the operational layers relevant to their responsibilities.

Simplify operational complexity without reducing operational visibility.

ARCHITECTURE

Built around the job card lifecycle

The platform architecture was centred on the job card, connecting operational workflows, technician activity, inventory and service states into a single execution system.

- Vehicle check-in - VIN or plate identification
- Customer verification and pre-inspection
- Job card created, services and parts attached
- Assigned to a bay, assigned to a technician
- Service execution with live state updates
- Inventory drawn down against the job card
- Checkout, invoice, reconciliation

BAY OPERATIONS

Bay operations & vehicle check-in

Optimising real-time check-in workflows.

Live bay monitoring

Real-time visibility into occupancy and operational states.

Operational state visibility

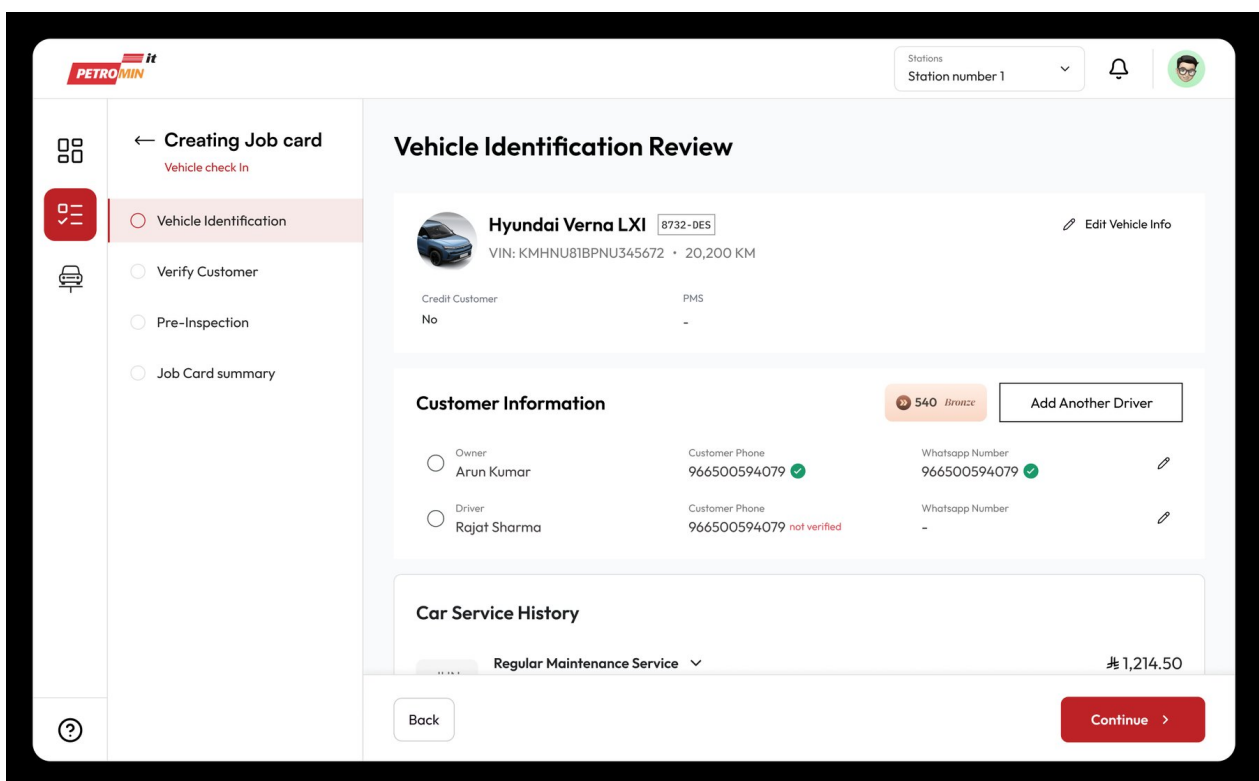
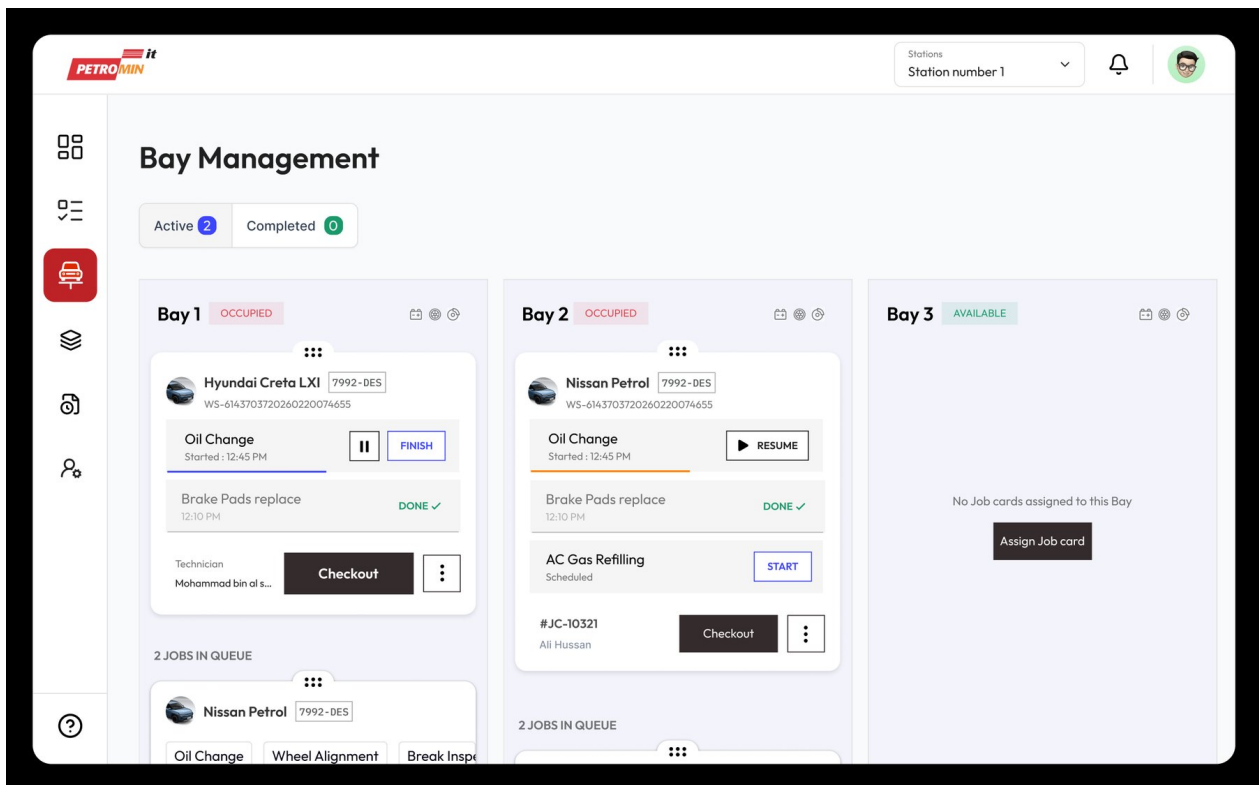
Service teams could instantly track vehicle progression across active bays.

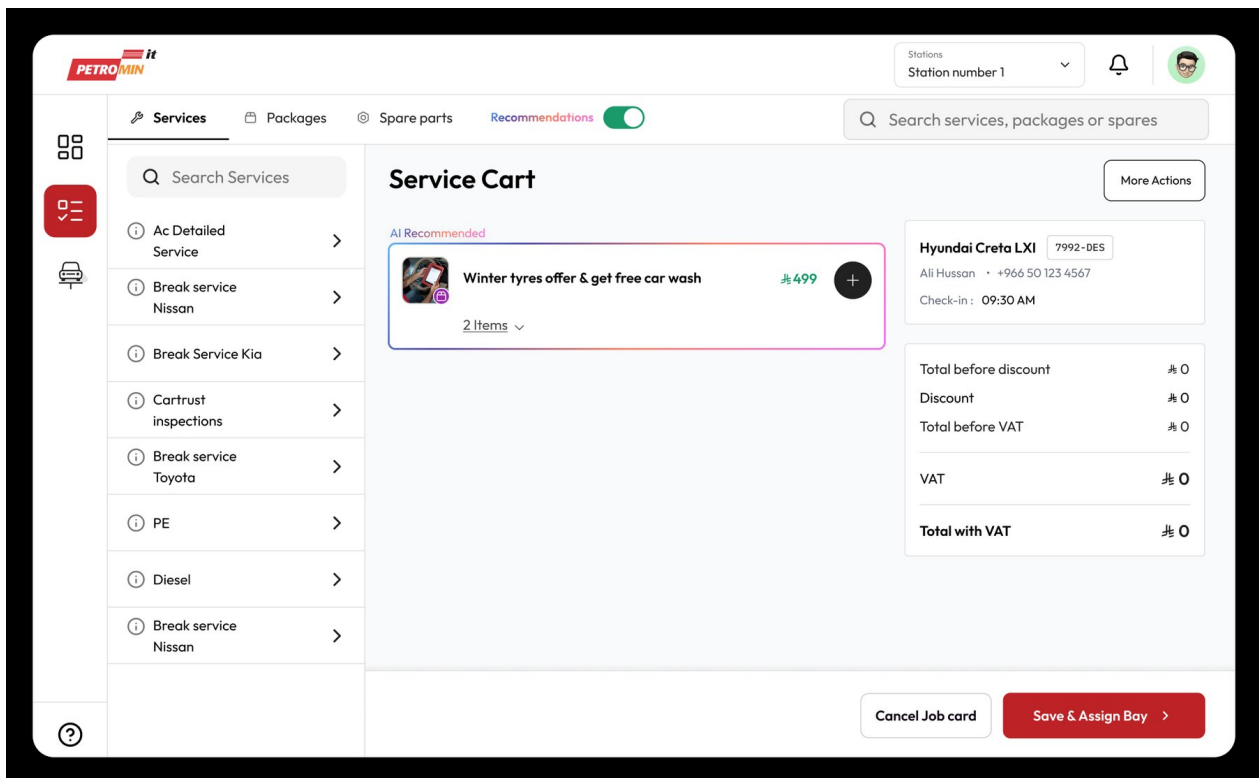
Quick vehicle identification

VIN or plate-based identification reduced manual entry effort.

Guided job creation

Service recommendations and contextual workflows accelerated check-in operations.





DESIGN SYSTEM

A scalable operational design system

Six modules and eight roles need one vocabulary, or every team invents its own.

Cards, tables, chips, action hierarchy, workflow panels.

Colour, operational state colours, typography, spacing.

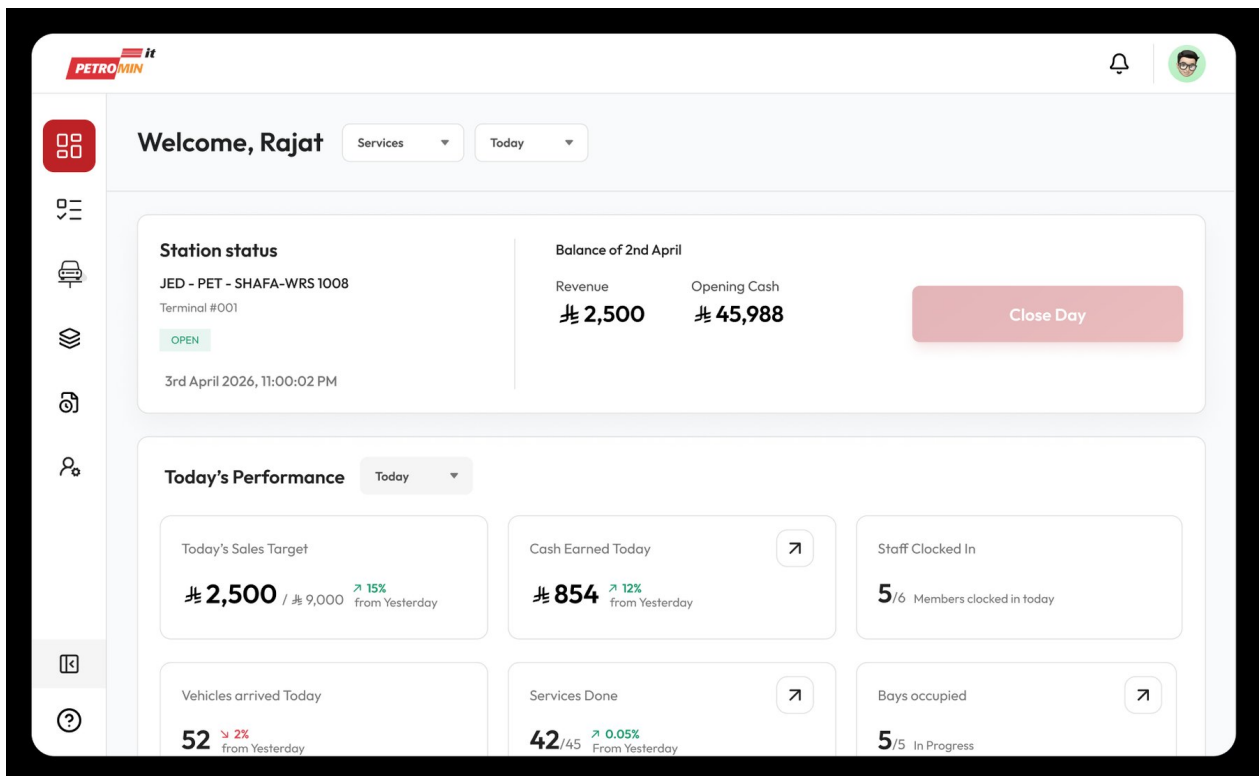
Cart widget, sidebar stepper, right drawer, reusable layouts.

In queue, in progress, on hold, scheduled, done, checked out. Six states that had to read identically on a technician's tablet, a manager's dashboard and a finance report - because the same job card appears in all three.

MANAGEMENT VIEW

Real-time operational visibility

Management teams gained centralised visibility into workshop operations, service throughput and operational performance - branch analytics, service KPIs, technician productivity, inventory insights, financial tracking.



The area manager's answer to "I only learn about problems after they happen"

CHALLENGES

Balancing operational complexity with usability

Depth without density

Eight roles need eight different amounts of information. Role-based complexity meant each person saw only their layer - the depth stayed in the system, out of the interface.

Speed on a shop floor

A technician with oily hands on a tablet is not a desk user. Scan-first interactions and large targets mattered more than any screen we could have drawn.

EXPECTED OUTCOMES

What the design is built to deliver

One technician-first ecosystem across technicians, service teams, finance, inventory and management - built for operational clarity, workflow efficiency and real-time visibility.

These are projections, not measured results. Each one is modelled from task-completion times recorded while watching a single user carry out that specific task on the legacy tools, compared against the same task in the new flow.

- Multi-role platform across eight responsibilities
- Real-time workflows on a live dashboard

- Tablet and web, matched to where the work happens
 - Workflow-centric UX built on operational states
 - Enterprise scalability across the station network
 - Service, inventory and finance in separate systems
 - Approvals queued behind busy hours
 - Visibility that arrived after the fact
 - Technicians switching tools mid-job
 - Inventory reconciled by hand
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TAKEAWAYS

Three things this taught me

Nobody was missing information - it all existed somewhere. The problem was never data, it was that no single view assembled it in time to act on.

Five roles described the same fracture from five sides. Interviewing across the hierarchy, not just the primary user, is what made the shape of it visible.

Choosing the job card as the architectural centre decided more than any screen did. Get the spine right and the modules hang off it; get it wrong and every screen argues with the next.

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